

Machine Learning Study Notes - Unit 3: Model Evaluation

Overfitting occurs when a model learns the training data too well, including its noise, and therefore performs poorly on unseen data. The standard remedy is to hold out a validation set and stop training when validation loss stops improving, a technique called early stopping.

Cross-validation splits the dataset into k folds; the model trains on $k-1$ folds and validates on the remaining fold, rotating k times. Five or ten folds are the most common choices.

Precision is the fraction of predicted positives that are truly positive.

Recall is the fraction of actual positives the model successfully finds.

The F1 score is the harmonic mean of precision and recall, used when classes are imbalanced and both error types matter.

Regularization discourages overly complex models. L1 regularization (lasso) adds the absolute value of weights to the loss and can drive weights exactly to zero, performing feature selection. L2 regularization (ridge) adds squared weights and shrinks them smoothly toward zero without eliminating them.

A confusion matrix tabulates predictions against true labels: true positives, false positives, true negatives, and false negatives. Accuracy is misleading on imbalanced data - a model predicting the majority class everywhere can score high accuracy while being useless.